

Quiz Night: a real-time pub quiz app

Cover

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Requirements and design

I analysed three existing quiz apps and derived requirements, prioritised by risk. The design uses WebSockets for real-time scoring; I justify this against polling with a latency estimate and include sequence diagrams for joining a game and answering.

Implementation

Implemented in Node.js with Socket.IO and PostgreSQL. All requirements are met, plus reconnection handling for dropped players. The code is modular with a clear separation between game logic and transport.

Testing and evaluation

Unit tests cover the game state machine (31 tests) and an automated load test ran 200 simulated players. I evaluate latency against my design estimate and discuss where it fell short.

Reflection

My summer placement at Fabrikam Games shaped how I planned sprints. I reflect on a poor early decision to store state in memory and how refactoring it cost two weeks, and I set out what I will do differently.